EE4047 – High Voltage Techniques I

Continuation to:

Turkish National Electric Power System and The Fundamentals of High Voltage Engineering

Week 2.2 – Thursday Lesson

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OUTLINE

Continuation to: 2 Turkish National Electric Power System

3 The Fundamentals of High Voltage Engineering

TURKISH NATIONAL ELECTRIC POWER SYSTEM

Turkish National Electric Power System Today

- Annual increase of the number and power of transformers in the transmission system

Year	400 kV		154 kV		66 kV		Total	
	Qt.	Power (MVA)	Qt.	Power (MVA)	Qt.	Power (MVA)	Qt.	Power (MVA)
2010	197	37.870	1.067	61.365	53	617	1.317	99.852
2011	203	39.620	1.105	64.470	49	568	1.357	104.658
2012	222	43.795	1.153	68.458	50	593	1.425	112.846
2013	245	48.540	1.212	73.123	48	573	1.505	122.236
2014	255	50.415	1.245	76.317	48	573	1.550	127.705
2015	282	56.665	1.302	81.365	42	521	1.628	138.951
2016	301	61.040	1.337	84.056	35	444	1.675	145.940
2017	327	66.378	1.392	95.972	29	431	1.750	163.181
2018	353	70.652	1.434	101.085	39	539	1.826	172.276
2019	370	75.102	1.480	105.722	39	536	1.889	181.360

2 TURKISH NATIONAL ELECTRIC POWER SYSTEM

Turkish National Electric Power System Today

- Turkish generation and transmission system is managed via 9 regional dispatching centers and coordinated by the National Dispatching Center.
- Power system operation is controlled by SCADA2 and Energy Management System EMS3 software.
- SCADA system includes 380 kV lines and power plants greater than 50 MW. System operator can manage every kind of system studies necessary for more quality, daily operating programs and system frequency control.

Generation and Transmission of electric energy

- The first public power station was put into service in 1882 in London (Holborn). Soon, a number of other public supplies for electricity followed in other countries.
- The early systems produced direct current (DC) at low-voltage and their service was limited to highly localized areas and they were mainly used for electric lighting.
- The limitations of DC transmission at low-voltage became readily apparent.
- After 1890, development of AC generator and transformer became more common and displaced earlier DC system.
- The first major AC power station was comission in 1890 at Deptford, UK, supplying power to central London, over a distance of 28 miles at 10 000 V.
- From the earliest days after the foundation of electricity, it was realized that to make full use of economic generation, the transmission network must be tailored to production with increased interconnection for pooling of generation in an integrated system.

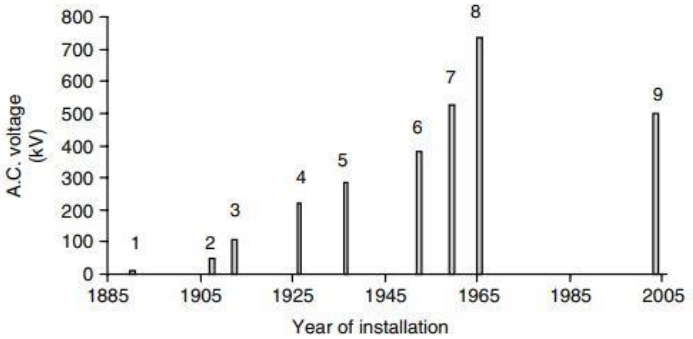
Generation and Transmission of electric energy

- Additionally, the development of hydroelectric power and the need to carry that power over long distances to the centres of consumption were recognized.
- Power transfer for large systems, whether in the context of interconnection of large systems or bulk transfers, led engineers to think in terms of high system voltages.
- The electric power (P) transmitted on an overhad AC line increases approximately with square of the system's operating voltage. For example, for a transmission line of surge impedance $Z_L = 250\Omega$, the power transfer capability is approximately $P = {V^2}/{Z_L}$, which for an overhead AC system, leading to the following results:

V (kV)	400	700	1000	1200	1500
P (MW)	640	2000	4000	5800	9000

Generation and Transmission of electric energy

- The rapidly increasing transmission voltage level in recent decades is a result of the growing demand for electrical energy, coupled with the development of large power stations at sites far remote from load centres and the need to transmit the energy over long distances to the centres.
- However, environmental concerns,have imposed limitations on system expansion resulting in the need to better utilize existing transmission systems.
- Although the majority of world's electric transmission is carried on AC systems, high voltage direct current (HVDC) provides an attractive alternative for bulk power transfer requiring long cables.



1	1890	10 kV	Deptford
2	1907	50 kV	Stadtwerke München
3	1912	110 kV	Lauchhammer – Riesa
4	1926	220 kV	N. Pennsylvania
5	1936	287 kV	Boulder Dam
6	1952	380 kV	Harspränget – Hallsberg
7	1959	525 kV	USSR
8	1965	735 kV	Manicouagan – Montreal
9	2003 (Est)	500 kV	Three Gorges (China)

Figure 1.1 *Major a.c. systems in chronological order of their installations*

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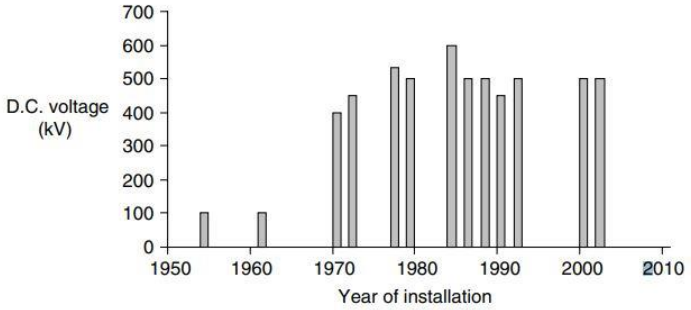


Figure 1.2 *Major d.c. systems in chronological order of their installations*

Generation and Transmission of electric energy

- Rather than having every power station or small group of power stations look after its own loads, tie lines connect power stations into an interconnected network over an entire country or a group of neighboring countries. These interconnections greatly improve the reliability and economy of the system and make available benefits fo diversity among the peaks of the component loads.
- The heavier the power to be transmitted and the longer the transmission distance, the higher the line voltage will have to be.
- This is determined by several technical and economical factors, including efficiency, voltage drop, system stability and number of circuits necessary for a secure supply of energy. The design, installation and operation of high-voltage equipment involve numerous technical challenges. For instance, a higher overhead line voltage involves larger seperations among power conductors and clearances to their tower structures and to ground. This entails a huge increase in the cost of the structure and of the system insulation in general.

Voltage Stresses

- Normal operating voltage does not severely stress the power system equipment's insulation and only in special circumstances, for example under pollution conditions, may operating voltage cause problems to external insulations. Nevertheless, the operating voltage determines the dimensions of the insulation which forms part of the generation, transmission and distribution equipment.
- The voltage stresses on power systems arise from various overvoltages which may be of external or internal origin.
- External overvoltages are associated with lightning discharges and are not dependent on the voltage of the systems.
- Internal overvoltages are generated by changes in the operating conditions of the systems such as switching operations, a fault on the system or fluctuations in the load or generation.

Voltage Stresses

- In designing the system's insulation, the two areas of specific importance are
  - (i) determination of the voltage stresses which the insulation must withstand,
  - (ii) Determination of the response of the insulation when subjected to these voltage stresses.
- Equipment insulation is stressed both by continuous normal voltage and by relatively slow dynamic and fast transient over-voltages.
- For insulation to withstand the various stresses, it has to be carefully designed, coordinated and properly tested.
- The test voltages have to exceed the normal rated voltages by some factor of safety.
- The lightning impulse test voltage is about 10 times the rated AC voltage for low-voltage equipment and about 3 for UHV equipment for the sake of system economy.

Voltage Stresses

- To achieve an economically and technically sound design for equipment insulation, adequate information needs to be acquired.
- This applies to equipment in power as well as electronics systems, regardless of the size.
- It applies equally to components and to devices.
- We should be able to calculate and measure system over-voltages. We should also be able to translate these voltage magnitudes and durations to electrical stresses across the vulnerable parts of equipment insulation and, accordingly, to dimension them.
- This involves computing and, in some cases, measuring electric fields.
- Additional necessary information includes the physics of breakdown of solid, liquid, and gaseous insulating materials, and the design, service performance, and testing of insulating components.

High voltage forms

- High voltage is present around us in a variety of forms.
- Lightning discharges are the only known natural form of high voltage.
- All the other forms of high voltage are human-made or synthetic to fulfill specific goals.
- There are three main purpose-oriented modes of human-made high voltage.

(i) The most well known mode is the use of high voltage in electric power transmission to avoid line currents. For a given power, a higher voltage requires smaller current and therefore smaller conductor cross-sectional area and lower conductor costs. Moreover, the current-dependent copper losses are reduced.

(ii) The second mode in which high voltage is utilized is charging objects under high voltage to develop an electrostatic force. Spray painting, cathode-ray tubes, particle accelerators and electrostatic precipitators that remove vast quantities of dust from flue gases of factories and power stations which would otherwise pollute the atmosphere can be given as examples.

- **High voltage forms**

(iii) The third mode of high voltage presence makes use of the ability of high voltage to initiate ionization in dielectric materials where energy is subsequently released in controlled quantities. Ignition in internal combustion engines, gas-discharge lamps, ozone generates that eliminate unpleasant smells from urban waste as it passes through sewage-treatment plants are some application examples.

- There are also other areas of high-voltage engineering, including the design of high voltage power supplies for various aerospace and ground applications, high-voltage, high-frequency sources for radio transmission; design, operation and testing problems for high voltage, extra high voltage and ultra high voltage overhead lines, cables and other equipment and their insulation coordination, various industrial, medical and other applications of gas discharge and many other high-voltage phenomena.
- Physics and chemistry also play prominent roles in high-voltage technology, in which the physical and chemical phenomena governing the electrical properties of insulating materials are of fundamental significance to high-voltage engineers.

Classification of Voltage

- The International Electrotechnical Commission (IEC Standards) defines high voltage as **being greater than 1000 V AC, or greater than 1200 V DC.**

Table 1.1 Standard Operating Voltages in Europe and the USA

Voltage class	Normal line voltage	
	Europe (50 Hz)	USA (60 Hz)
Low (LV)	220/240	120 (1-ph)
	380/415	208
	650	600
	1000	—
	kV	kV
Medium (MV)	—	2.4
	5	6.9
	11	12.47
	22	23
	33	34.5
	66	69.0
High (HV)	110	115
	132	138
	156	161
	220	230
Extra high (EHV)	275	287
	380	345
	400	500
	800	765
Ultra high (UHV)	1150	

HVDC

- Commonly used for energy transfer from offshore areas using submarine cables.
- In1954, 96 km transmission line from Sweden to Gotland Island.
- One conductor capable of carrying power up to 30 MW with 150 kV.
- Today commonly used in Nordic countries and US.
- With advances in high voltage high current semiconductor components, HVDC became a more feasible alternative compared to AC.

HV Components

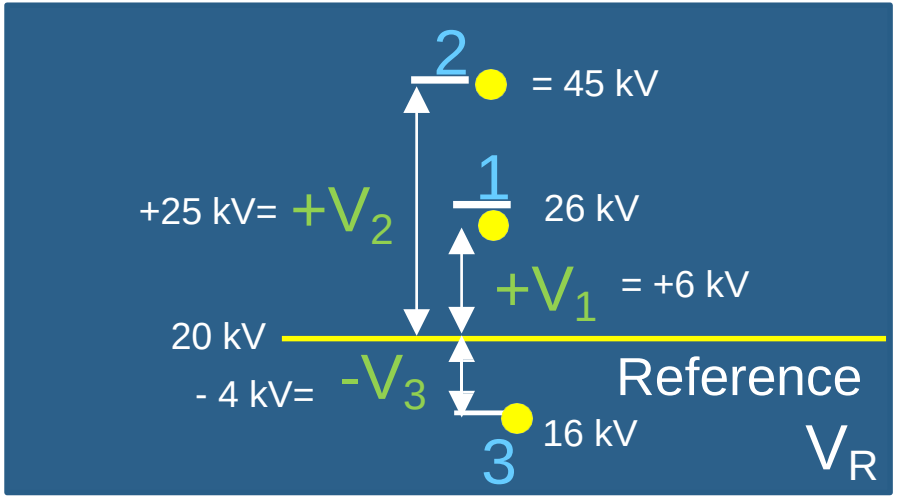
- Main equipment: synchronous generators, power transformers, circuit breakers, sectionalizers, towers, isolators, cables and busses
- Measurement and protection equipment: Current and voltage transformers, voltage dividers, surge arresters, spark gap, corona rings, relays.
- Control Equipment: Relays, circuit breakers, control and management systems.
- Voltage Regulation Equipment: Series-parallel connected reactors, series-parallel connected capacitors

HV Concepts

Potential, V:

The electrical value of a point according to a reference.

Unit: Volt

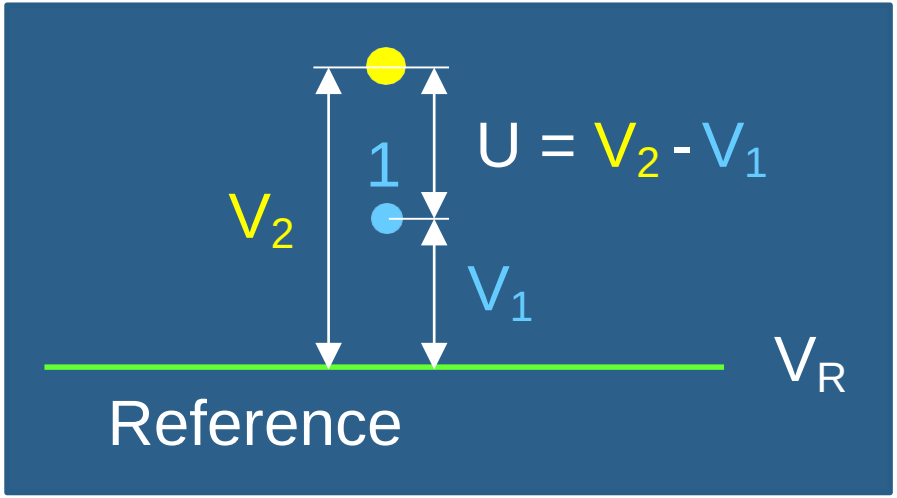


HV Concepts

Voltage, U:

The potential difference.

Unit: Volt

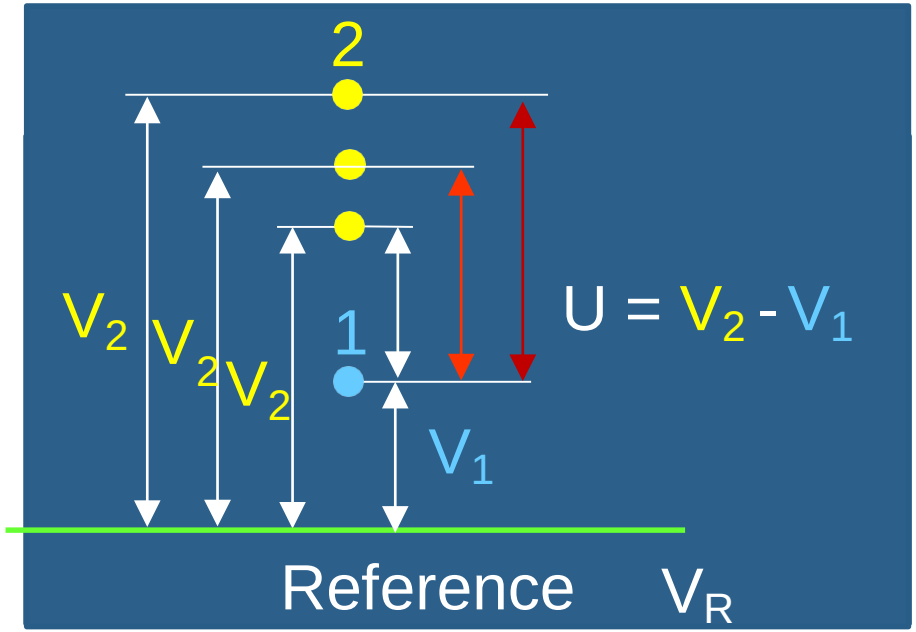


HV Concepts

Voltage, U:

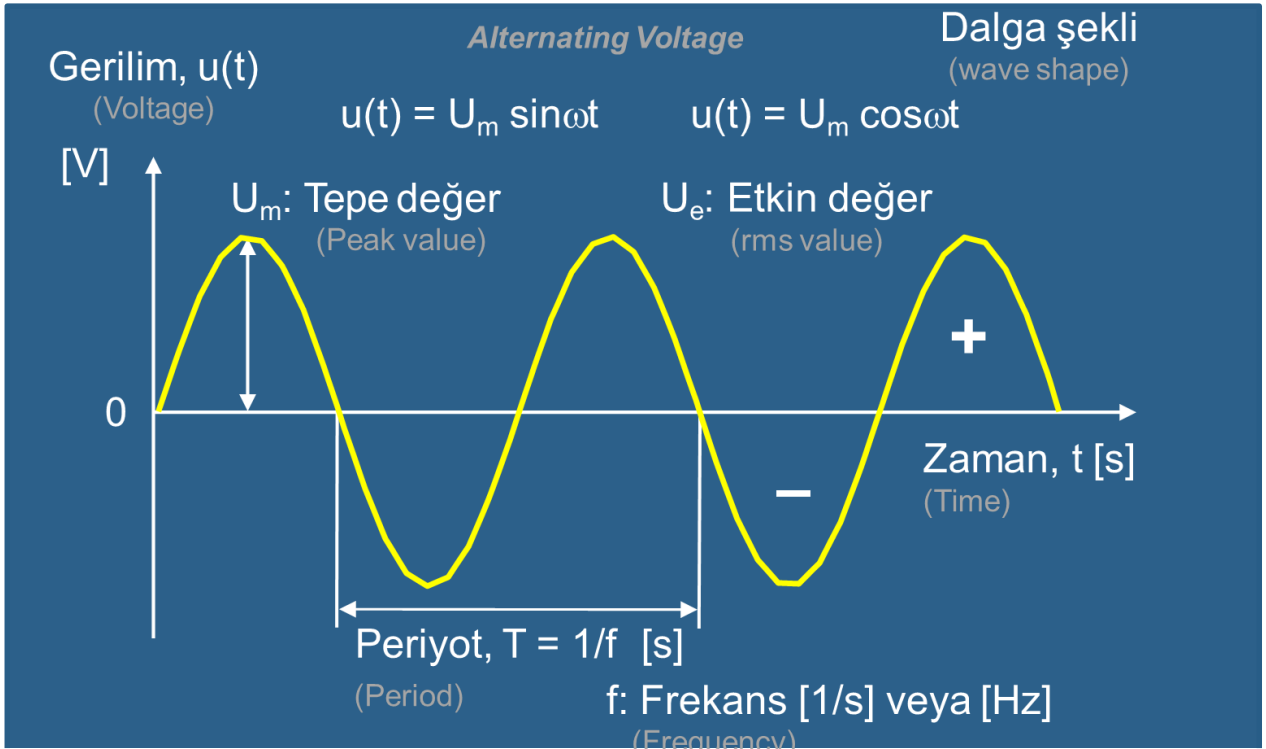
The potential difference.

Unit: Volt



HV Concepts

Alternating Voltage:



END OF THE LESSON

Any questions?

Check out the Power Engineering playlist with 32 videos on Youtube:

https://www.youtube.com/watch?v=JwRTpWZReJk&list=PLLZI7WaEmZi3F134eDujF5ZNGxA1E0dkm